Car Showroom Management System

1. Introduction to the Working of the System:

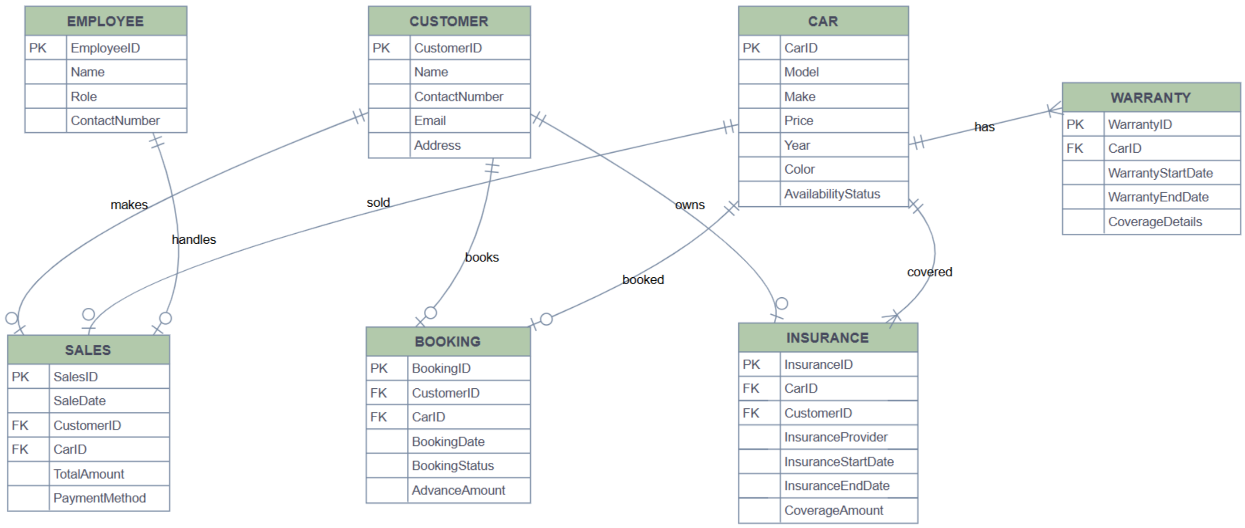
The **Car Showroom Management System** helps manage car showroom operations efficiently by automating key tasks like customer management, car inventory, and sales tracking.

- **Key Features of the System**:
  1. Maintains detailed records of customers and their interactions with the showroom.
  2. Manages car inventory with attributes like model, price, availability, and specifications.
  3. Processes sales transactions, including customer details, payment methods, and delivery schedules.
  4. Generates reports based on sales, inventory, and other essential metrics to support decision-making.

- **Database Design**:
  1. Developed using both the **top-down** and **bottom-up** approaches.
  2. Includes an **ER Diagram (ERD)** to illustrate relationships between entities like Customers, Cars, and Sales.
  3. Tables are defined with attributes, constraints, and relational keys for maintaining data integrity.

- **Advanced Functionalities**:
  1. **Custom Views**: Specific views are created for simplified access to the most important data.
  2. **Query-Based Reports**: Queries provide insights such as customer purchases, car availability, and overall performance.
- **Purpose**:
  The system aims to simplify workflows, ensure accurate record-keeping, and improve overall management for car showrooms, leading to enhanced customer service and business growth.

2. ERD Diagram:



<u>3. Construction of the Relational Schema</u> :

Top-Down approach :

1. **Customer** (<u>CustomerID</u>, Name, ContactNumber, Email, Address)

2. **Car** (<u>CarID</u>), Model, Make, Price, Year, Color, AvailabilityStatus

3. **Sales** (<u>SalesID</u>, SaleDate, **CustomerID** (FK referencing Customer), **CarID** (FK referencing Car), TotalAmount, PaymentMethod)

4. **Employee** (<u>EmployeeID</u>, Name, Role, ContactNumber)

5. **Insurance** (<u>InsuranceID</u>, **CarID** (FK referencing Car), **CustomerID** (FK referencing Customer), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount)

6. **Booking** (BookingID, **CustomerID** (FK referencing Customer), **CarID** (FK referencing Car), BookingDate, BookingStatus, AdvanceAmount)

7. **Warranty** (WarrantyID, **CarID** (FK referencing Car), WarrantyStartDate, WarrantyEndDate, CoverageDetails)

Bottom-Up Approach:

First Normal Form (1NF)

The schema is already in 1NF because each field contains atomic values, and there are no repeating groups or multiple values in any field. The data is organized into separate tables with unique primary keys.

Relations in 1NF

1. Customer Table
Customer(CustomerID, CustomerName, CustomerContactInfo)

2. Car Table
Car(CarID, CarModel, CarMake, CarPrice, CarYear, CarColor, CarAvailabilityStatus)

3. Sales Table
Sales(SalesID, CustomerID, CarID(FK), SaleDate, TotalAmount, PaymentMethod)

4. Employee Table
Employee(EmployeeID, EmployeeName, EmployeeContactInfo, EmployeeRole)

5. Insurance Table
Insurance(InsuranceID, CarID(FK), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount)

6. Booking Table
Booking(BookingID, CustomerID(FK), CarID(FK), BookingDate, BookingStatus, AdvanceAmount)

7. Warranty Table
Warranty(WarrantyID, CarID(FK), WarrantyStartDate, WarrantyEndDate, WarrantyCoverageDetails)

Second Normal Form (2NF)

The schema satisfies 2NF as it is in 1NF and there are no partial dependencies

Third Normal Form (3NF)

The schema satisfies 3NF as it is in 2NF and there are no transitive dependencies.

4.Description of Tables:

Customer:

Attribute	Data Type	Size	Constraints
CustomerID	NUMBER		Primary Key
Name	VARCHAR2	20	Not Null
ContactNumber	VARCHAR2	20	Unique
Email	VARCHAR2	20	Unique
Address	VARCHAR2	20	

Car Table:

Attribute	Data Type	Size	Constraints
CarID	NUMBER		Primary Key
Model	VARCHAR2	20	Not Null
Make	VARCHAR2	20	Not Null
Price	NUMBER	(10, 2)	Not Null
Year	NUMBER		Not Null
Color	VARCHAR2	20	Not Null
AvailabilityStatus	VARCHAR2	20	CHECK (AvailabilityStatus IN ('Available', 'Sold'))

Attribute	Data Type	Size	Constraints
SalesID	NUMBER		Primary Key
SaleDate	DATE		Not Null
CustomerID	NUMBER		Foreign Key References Customer(CustomerID)
CarID	NUMBER		Foreign Key References Car(CarID)
TotalAmount	NUMBER	(10, 2)	Not Null
PaymentMethod	VARCHAR2	20	CHECK (PaymentMethod IN ('Cash', 'Card', 'Financing'))

Employee Table:

Attribute	Data Type	Size	Constraints
EmployeeID	NUMBER		Primary Key
Name	VARCHAR2	20	Not Null
Role	VARCHAR2	20	Not Null
ContactNumber	VARCHAR2	20	Unique

Insurance Table:

Attribute	Data Type	Size	Constraints
InsuranceID	NUMBER		Primary Key
CarID	NUMBER		Foreign Key References Car(CarID)
CustomerID	NUMBER		Foreign Key References Customer(CustomerID)
InsuranceProvider	VARCHAR2	20	Not Null
InsuranceStartDate	DATE		Not Null
InsuranceEndDate	DATE		Not Null
CoverageAmount	NUMBER	(10, 2)	Not Null

Booking Table:

Attribute	Data Type	Size	Constraints
BookingID	NUMBER		Primary Key
CustomerID	NUMBER		Foreign Key References Customer(CustomerID)
CarID	NUMBER		Foreign Key References Car(CarID)
BookingDate	DATE		Not Null
BookingStatus	VARCHAR2	20	CHECK (BookingStatus IN ('Pending', 'Confirmed', 'Cancelled'))
AdvanceAmount	NUMBER	(10, 2)	

Warranty Table:

Attribute	Data Type	Size	Constraints
WarrantyID	NUMBER		Primary Key
CarID	NUMBER		Foreign Key References Car(CarID)
WarrantyStartDate	DATE		Not Null
WarrantyEndDate	DATE		Not Null
CoverageDetails	VARCHAR2	20	

5.Create Table Statements:

Customer:

```
CREATE TABLE Customer (
    CustomerID NUMBER PRIMARY KEY,
    Name VARCHAR2(20) NOT NULL,
    ContactNumber VARCHAR2(20) UNIQUE,
    Email VARCHAR2(20) UNIQUE,
    Address VARCHAR2(20)
```

);

Car Table:

```
CREATE TABLE Car (
    CarID NUMBER PRIMARY KEY,
    Model VARCHAR2(20) NOT NULL,
    Make VARCHAR2(20) NOT NULL,
    Price NUMBER(10, 2) NOT NULL,
    Year NUMBER NOT NULL,
    Color VARCHAR2(20) NOT NULL,
    AvailabilityStatus VARCHAR2(20) CHECK (AvailabilityStatus IN ('Available', 'Sold'))
);
```

Sales Table:

```
CREATE TABLE Sales (
    SalesID NUMBER PRIMARY KEY,
    SaleDate DATE NOT NULL,
    CustomerID NUMBER,
    CarID NUMBER,
    TotalAmount NUMBER(10, 2) NOT NULL,
    PaymentMethod VARCHAR2(20) CHECK (PaymentMethod IN ('Cash', 'Card', 'Financing')),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (CarID) REFERENCES Car(CarID)
);
```

Employee Table:

```
CREATE TABLE Employee (
```

```
    EmployeeID NUMBER PRIMARY KEY,
    Name VARCHAR2(20) NOT NULL,
    Role VARCHAR2(20) NOT NULL,
    ContactNumber VARCHAR2(20) UNIQUE
);
```

Insurance Table:

```
CREATE TABLE Insurance (
    InsuranceID NUMBER PRIMARY KEY,
    CarID NUMBER,
    CustomerID NUMBER,
    InsuranceProvider VARCHAR2(20) NOT NULL,
    InsuranceStartDate DATE NOT NULL,
    InsuranceEndDate DATE NOT NULL,
    CoverageAmount NUMBER(10, 2) NOT NULL,
    FOREIGN KEY (CarID) REFERENCES Car(CarID),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
);
```

Booking Table:

```
CREATE TABLE Booking (
    BookingID NUMBER PRIMARY KEY,
    CustomerID NUMBER,
    CarID NUMBER,
    BookingDate DATE NOT NULL,
    BookingStatus VARCHAR2(20) CHECK (BookingStatus IN ('Pending', 'Confirmed', 'Cancelled')),
    AdvanceAmount NUMBER(10, 2),
```

```
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (CarID) REFERENCES Car(CarID)
);
```

Warranty Table:

```
CREATE TABLE Warranty (
    WarrantyID NUMBER PRIMARY KEY,
    CarID NUMBER,
    WarrantyStartDate DATE NOT NULL,
    WarrantyEndDate DATE NOT NULL,
    CoverageDetails VARCHAR2(20),
    FOREIGN KEY (CarID) REFERENCES Car(CarID) );
```

6. Desing of at least two VIEWS, that you feel are the most important.

1. Sales_info_View:

```
CREATE VIEW Sales_Info_View AS
SELECT
    S.SalesID,
    S.SaleDate,
    S.TotalAmount,
    C.Name AS CustomerName,
    C.ContactNumber AS CustomerContact,
    Ca.Model AS CarModel,
    Ca.Make AS CarMake
FROM
    Sales S
JOIN
    Customer C ON S.CustomerID = C.CustomerID
```

JOIN

Car Ca ON S.CarID = Ca.CarID;

SALESID	SALEDATE	TOTALAMOUNT	CUSTOMERNAME	CUSTOMERCONTACT	CARMODEL	CARMAKE
1	01/15/2024	2500000	Ali Khan	03331234567	Corolla	Toyota
2	01/18/2024	3000000	Sara Ahmed	03339876543	Civic	Honda

2 rows returned in 0.01 seconds Download

2.Car_Warranty_View:

CREATE VIEW Car_Warranty_View AS

SELECT

Ca.CarID,

Ca.Model AS CarModel,

Ca.Make AS CarMake,

Ca.Color AS CarColor,

W.WarrantyStartDate,

W.WarrantyEndDate,

W.CoverageDetails

FROM

Car Ca

JOIN

Warranty W ON Ca.CarID = W.CarID;

CARID	CARMODEL	CARMAKE	CARCOLOR	WARRANTYSTARTDATE	WARRANTYENDDATE	COVERAGEDETAILS
1	Corolla	Toyota	Silver	01/01/2024	01/01/2027	3 Years Warranty
2	Civic	Honda	Black	01/10/2024	01/10/2027	3 Years Warranty

2 rows returned in 0.00 seconds Download

7. SELECT statements:

1.Report on Car Availability:

```
SELECT
    Ca.Model AS CarModel,
    Ca.Make AS CarMake,
    Ca.Color AS CarColor,
    Ca.AvailabilityStatus,
    (SELECT COUNT(*) FROM Sales S WHERE S.CarID = Ca.CarID) AS CarsSold
FROM
    Car Ca
ORDER BY
    Ca.Make, Ca.Model;
```

Results Explain Describe Saved SQL History

CARMODEL	CARMAKE	CARCOLOR	AVAILABILITYSTATUS	CARSSOLD
Civic	Honda	Black	Available	1
Corolla	Toyota	Silver	Available	1

2 rows returned in 0.01 seconds Download

2.Report on Insurance Expiry:

```
SELECT
    Ca.Model AS CarModel,
    Ca.Make AS CarMake,
    Insh.InsuranceProvider,
    Insh.InsuranceEndDate,
    (SELECT AVG(CoverageAmount) FROM Insurance WHERE CarID = Ca.CarID) AS AverageCoverageAmount
FROM
    Insurance Insh
JOIN
    Car Ca ON Insh.CarID = Ca.CarID
WHERE
    Insh.InsuranceEndDate BETWEEN SYSDATE AND SYSDATE + 30
ORDER BY
    Insh.InsuranceEndDate;
```

Results Explain Describe Saved SQL History

CARMODEL	CARMAKE	INSURANCEPROVIDER	INSURANCEENDDATE	AVERAGECOVERAGEAMOUNT
Corolla	Toyota	Jubilee Life	01/01/2025	1000000
Civic	Honda	EFU Life	01/10/2025	1200000

2 rows returned in 0.01 seconds Download

3. Report on Cars Purchased by Each Customer:

```
SELECT
    C.Name AS CustomerName,
    Ca.Model AS CarModel,
    Ca.Make AS CarMake,
    S.TotalAmount,
    (SELECT MAX(S1.TotalAmount) FROM Sales S1 WHERE S1.CustomerID = C.CustomerID) AS MaxAmountSpent
FROM
    Sales S
JOIN
    Customer C ON S.CustomerID = C.CustomerID
JOIN
    Car Ca ON S.CarID = Ca.CarID
ORDER BY
    CustomerName, Ca.Model;
```

Results Explain Describe Saved SQL History

CUSTOMERNAME	CARMODEL	CARMAKE	TOTALAMOUNT	MAXAMOUNTSPENT
Ali Khan	Corolla	Toyota	2500000	2500000
Sara Ahmed	Civic	Honda	3000000	3000000

2 rows returned in 0.01 seconds   Download

4. Report on Total Sales Amount by Car Model:

```
SELECT
    Ca.Model AS CarModel,
    Ca.Make AS CarMake,
    SUM(S.TotalAmount) AS TotalSalesAmount
FROM
    Sales S
JOIN
    Car Ca ON S.CarID = Ca.CarID
GROUP BY
    Ca.Model, Ca.Make
ORDER BY
    TotalSalesAmount DESC;
```

Results Explain **Describe** Saved SQL History

CARMODEL	CARMAKE	TOTALSALESAMOUNT
Civic	Honda	3000000
Corolla	Toyota	2500000

2 rows returned in 0.00 seconds   Download

5. **Report on Average Car Price by Car Make:**

```
SELECT
    Ca.Make AS CarMake,
    AVG(Ca.Price) AS AvgCarPrice
FROM
    Car Ca
GROUP BY
    Ca.Make
ORDER BY
    AvgCarPrice DESC;
```

Results Explain Describe Saved SQL History

CARMAKE	AVGCARPRICE
Honda	3000000
Toyota	2500000

2 rows returned in 0.00 seconds Download